BLOOM PRODUCT DATA CHALLENGE

Diagnosing the week 2–4 engagement drop

Analysis of ~2,500 Bloom members · May 2025 – Feb 2026

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1. FRAMING

The week 2–4 engagement drop is an early exercise habit problem

Point of view: the drop is mainly an early exercise habit formation problem. Members who fail to reach at least 2 of 3 prescribed sessions early are less likely to remain meaningfully engaged by week 4.

Recommendation: redesign onboarding as a weeks 0–4 habit-formation journey, with contextual AI coach support for members showing early low-adherence signals.

Target behavior

Help more members build and sustain the 2-of-3 weekly therapy routine early.

1. FRAMING

Primary metric: Week 4 meaningful exercise engagement

What we measure and why

PRIMARY METRIC

Week 4 Meaningful Engagement

% of the initial cohort completing ≥2 of 3 prescribed weekly exercise sessions by week 4

DENOMINATOR

Initial cohort

ENGAGEMENT THRESHOLD

Weekly adherence ≥ 66.7%

This makes engagement a therapy-adherence outcome, not just an app-activity outcome.

Tradeoff: more app opens, pushes, or AI messages are useful only if they help members complete the prescribed therapy routine.

WEEK 4 BASELINE

47.2%

of the initial cohort is still completing at least 2 of 3 prescribed weekly exercise sessions by week 4.

WHY THIS METRIC

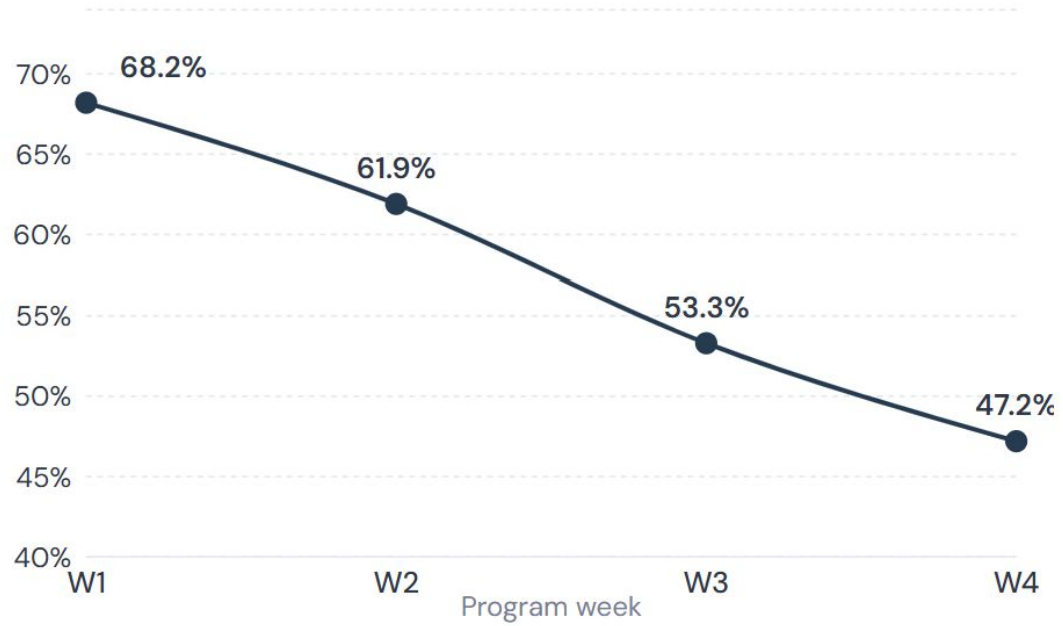
This metric combines retention and meaningful therapy adherence into one outcome tied to clinical behavior.

Not optimizing directly

- push received
- raw event volume
- app opens alone
- AI messages sent
- session duration alone
- short-term pain alone

2. DIAGNOSE THE DROP-OFF

The drop happens between weeks 2 and 4 in retention of the minimum exercise routine



61.9%

Week 2: initial cohort with adherence ≥66.7%

47.2%

Week 4: initial cohort with adherence ≥66.7%

This is a drop in the therapeutic behavior we care about: completing prescribed exercise sessions.

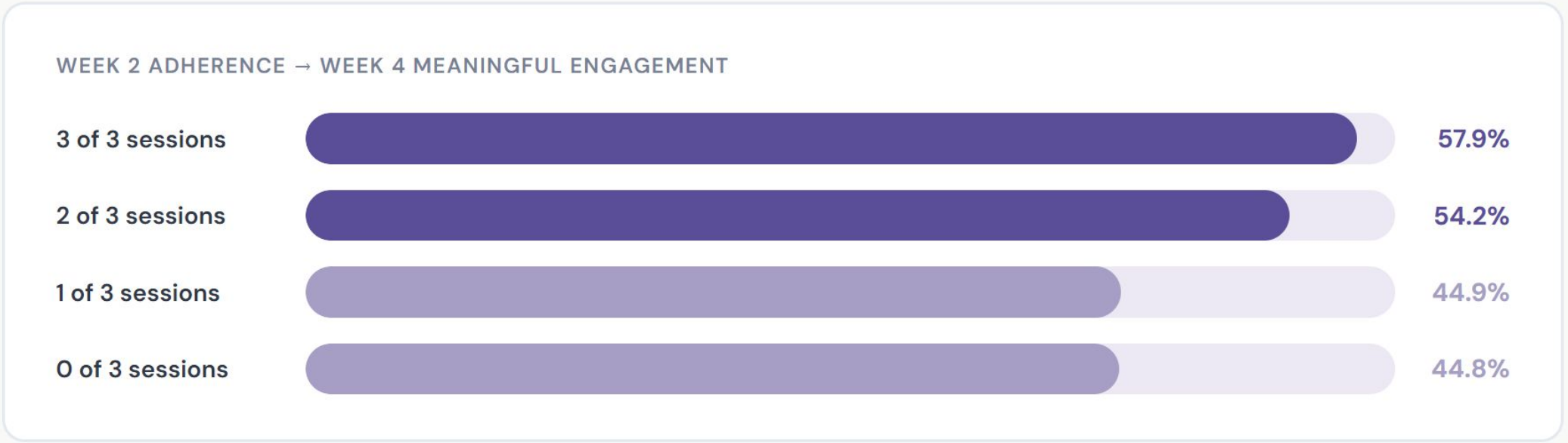
Among observed members, average adherence declines only modestly: 63.3% to 61.3%.

> The issue is not mainly lower adherence among active members. It is fewer original members sustaining the 2-of-3 weekly routine by week 4.

3. EARLY ADHERENCE SIGNAL

Early adherence is the clearest predictor of week 4 engagement

Members completing 0–1 of 3 sessions in week 2 are the highest-risk behavioral group.



Takeaway: the threshold that matters is reaching at least 2 of 3 sessions early.

Behavioral signals support the early habit-formation diagnosis

What declines after week 2, and what remains correlational

Declining Signals (Week 2 → 4)

- App opens per user: **5.5 → 3.5**
- Non-passive events per active user: **11.1 → 8.4**
- Push open rate: **52% → 42%**
- Missed-session events per user: **1.72 → 1.82**

AI Coach Signal

Early AI coach exposure is associated with higher week 4 engagement:

- 0 messages: **41.0%**
- 1–2 messages: **49.8%**
- 3–5 messages: **56.4%**

Correlational, not causal

BEHAVIORAL READ

App opens and completed exercises both decline, so the issue looks broader than in-session friction. Members are losing the weekly routine.

CAUSAL CAUTION

Higher push volume was not clearly associated with lower engagement. AI coach exposure may reflect selection bias, so it should be tested as contextual onboarding support rather than as a standalone bet.

5. ALTERNATIVE EXPLANATIONS CHECKED

The drop is not clearly explained by clinical deterioration, satisfaction, age, condition, or seasonality

Hypothesis	Evidence	Conclusion
Clinical experience deteriorates	Pain reduction stable; completed-session duration stable around 18 min	Not supported
Satisfaction drives churn	Satisfaction differences are small	Weak signal
Age explains the drop	Week 4 engagement is broadly stable across age bands	Not supported
Seasonality explains the drop	Early exercise completion is broadly stable across signup months	Not supported
Condition as sole explanation	Patterns vary by condition, but no single condition explains the overall drop	Secondary / tailoring signal
Region as primary driver	Week 4 engagement is similar across regions, around 46–49%	Weak / secondary

0.8
Average immediate pain reduction per completed session

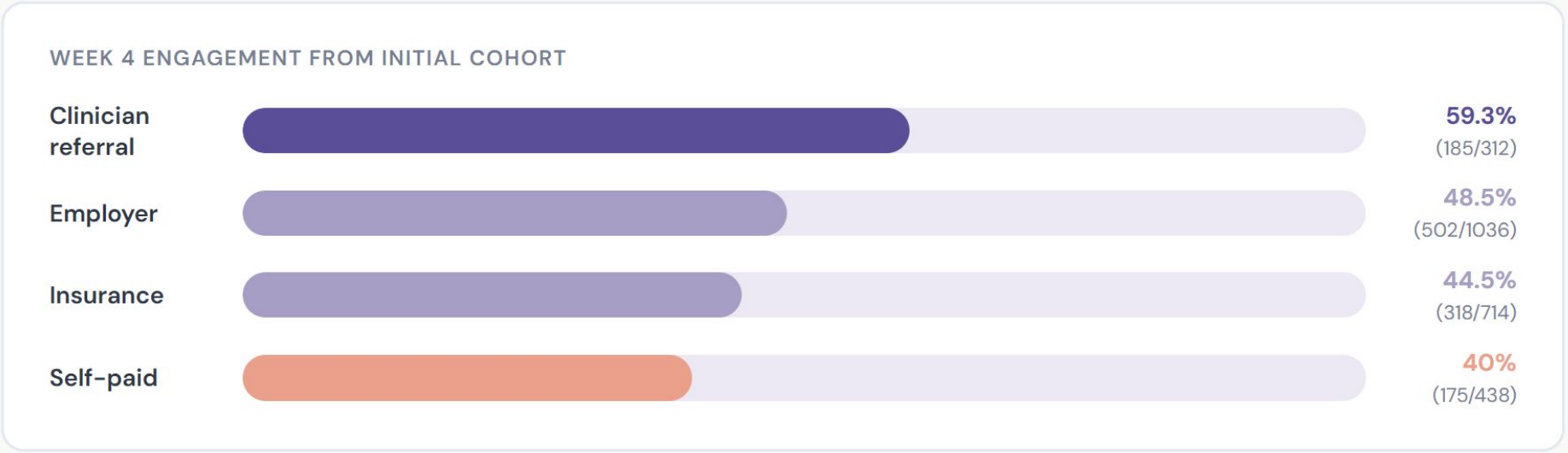
1.35
Pain reduction from first to last observed session among completers

These checks do not mean clinical factors never matter. They mean the observed drop is better explained by early habit formation than by deterioration in the completed-session experience.

6. SEGMENTATION

Referral source is the clearest 'for whom' signal

Self-paid members have the lowest week 4 engagement; clinician-referred members outperform by 19pp.



68.6%

Clinician-referred members who maintain W2→W4 engagement

47.1%

Self-paid members who maintain W2→W4 engagement

Interpretation: Referral source likely reflects **trust, motivation, and expectation-setting**. I would not interpret this as a payment effect without payer / coverage data. The product may need to replicate some of the clarity and accountability clinician-referred members already bring, especially for self-paid members.

7. RECOMMENDATION

Recommendation: redesign onboarding as a weeks 0–4 habit-formation journey

Redesigned onboarding is the main bet, with proactive AI coach support embedded in the early journey.

The goal is not more onboarding screens or basic first-use education; it is helping members convert an initial start into a consistent weekly therapy routine.

Why onboarding

- Intervenes before the week 4 drop is fully visible
- Targets the 2-of-3 weekly session routine
- Scales across referral sources and clinical segments

Why not AI coach alone

- AI exposure is associated with higher engagement
- But this may reflect selection bias
- Better tested as contextual onboarding support

Why not clinician outreach first

- Clinician referral is a strong signal
- But it does not prove human outreach caused the outcome
- Less scalable as a broad first intervention
- Test whether onboarding can replicate trust and accountability

What "Redesigned Onboarding" Means in Practice

Clear weekly plan · First-session support · Week 1/2 milestones · Progress feedback · Week 2 recap · Rescue flows for low adherence · Contextual AI coach for members showing early drop signals · Value communication tailored to referral source

8. EXPERIMENT DESIGN & NEXT STEPS

A/B test the weeks 0–4 habit-formation onboarding journey

Experiment design

Population: new Bloom members

Randomization: member-level

Stratification: referral source, condition, plan length, baseline pain

Control: current onboarding

Treatment: weeks 0–4 habit-formation onboarding journey

47.2%
Observed baseline

+5–6pp
Initial MDE target

Primary: Week 4 meaningful exercise engagement retention

Secondary: Week 2 adherence, completed/missed sessions, churn, app opens, AI interactions

Guardrails: Pain, satisfaction, support tickets, notification/AI fatigue

What I would want next

- Confirm operational definition of `churned_this_week`.
- Add onboarding / education / assessment funnel events and time to first exercise.
- Capture churn / missed-session reasons.
- Add payer / coverage data to separate referral effects from payment model.
- Instrument AI coach downstream impact within 24–48h.

> The biggest opportunity is helping more members reach and sustain the weekly therapy routine in the first 4 weeks.